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Webinar: #1

SOLUTION OF LINEAR SYSTEM



- i The rank of a matrix **A** is number of pivot elements in REF(**A**).
- ii Consider a system of linear equation expressed as a matrix equation **Ax = b**

The system is consistent if rank (**A|b**) = rank (**A**).

Suppose if the system is consistent

- ▶ It has unique solution if rank (**A|b**) = rank (**A**) = number of columns of **A**.
- ▶ It has infinitely many solutions if rank (**A|b**) = rank (**A**) < number of columns of **A**.

PROBLEM 1

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You are given a system of three linear equations as shown below:

$$\alpha x + 2y + 6z = b_1$$

$$\alpha x + \alpha y + 5z = b_2$$

$$\alpha x + \alpha y + \alpha z = b_3$$

In each of the equations, you can observe the presence of an unknown constant α . Derive the three possible values of α for which the given linear system will fail to have three pivots. Provide a valid reason in each case.

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Consider the coefficient matrix

$$\mathbf{A} = \begin{bmatrix} \alpha & 2 & 6 \\ \alpha & \alpha & 5 \\ \alpha & \alpha & \alpha \end{bmatrix}.$$

To convert it into Row Echelon Form (REF), perform row operations:

$$\mathbf{A} \xrightarrow{R_2 \leftarrow R_2 - R_1, R_3 \leftarrow R_3 - R_1} \begin{bmatrix} \alpha & 2 & 6 \\ 0 & \alpha - 2 & -1 \\ 0 & \alpha - 2 & \alpha - 6 \end{bmatrix}.$$

Continue row reduction:

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$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} \alpha & 2 & 6 \\ 0 & \alpha - 2 & -1 \\ 0 & 0 & \alpha - 5 \end{bmatrix}.$$

Hence, the REF of **A** is

$$\begin{bmatrix} \alpha & 2 & 6 \\ 0 & \alpha - 2 & -1 \\ 0 & 0 & \alpha - 5 \end{bmatrix}.$$

Clearly, **A** will not have 3 pivots if

$$\alpha = 0, \quad \text{or} \quad \alpha - 2 = 0, \quad \text{or} \quad \alpha - 5 = 0.$$

$$\boxed{\alpha = 0, 2, 5}.$$

VECTOR SPACES, SUBSPACES AND DIMENSION

- Let V be a vector space over F and $W \subset V$. W is a subspace over F if
 - $\mathbf{0} \in W$ and
 - $\alpha\omega_1 + \beta\omega_2 \in W \quad \forall \omega_1, \omega_2 \in W$ and $\alpha, \beta \in F$.
- The linear span of a set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is the set

$$LS(S) = \left\{ \sum_{i=1}^n \alpha_i \mathbf{v}_i \mid \alpha_i \in F, \forall i = 1, 2, \dots, n \right\}.$$

- A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is linearly independent if

$$\sum_{i=1}^n \alpha_i \mathbf{v}_i = \mathbf{0}$$

has $\alpha_i = 0$ as the only solution for all $i = 1, 2, \dots, n$. If at least one of the $\alpha_i \neq 0$, then it is said to be linearly dependent.

- A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is called a basis of a vector space V over F if S is linearly independent and spans V . An equivalent definition is that S is the maximum number of linearly independent elements in V . A vector space can have multiple bases.
- Number of basis vectors is called as dimension.

PROBLEM 2



$$\text{Let } \mathbf{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

- (a) Using elementary row operations, write the matrix in its REF.
- (b) Let $\mathbf{V}$ be a vector subspace spanned by the columns of matrix $\mathbf{A}$. Give a set of vectors that form the basis for the vector space $\mathbf{V}$. What is the number of vectors in this basis?
- (c) Let $\mathbf{V}$ be a vector subspace spanned by vectors $\mathbf{x}$, such that $\mathbf{Ax} = \mathbf{0}$. Give a set of vectors that form the basis for the vector space $\mathbf{V}$. What is the number of vectors in this basis?
- (d) Give the set of linearly independent rows of $\mathbf{A}$. What is the number of vectors in this set?

SOLUTION



(a)

$$\mathbf{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ -3 & 6 & -1 & 1 & -7 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 + 3 * R_1, R_3 \leftarrow R_3 - 2 * R_1, R_2 \leftarrow R_2 / 5, R_3 \leftarrow R_3 - R_2, R_1 \leftarrow R_1 - 2 * R_2}$$

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

NOTE: REF is not unique.

(b) column number 1 and column number 3 have pivot.
Basis of column space is:

$$\begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$$

Rank of col space = 2

(c) From REF, the null space is:

$$\begin{bmatrix} x \\ r \\ y \\ s \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} r + \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} s + \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} t$$

Rank of null space = 3

(d) From REF, row 1 and row 2 form basis of row space.

Rank of row space = 2

PROBLEM 3

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Suppose $\mathbf{A} = [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4]$, where $\mathbf{C}_m \in \mathbb{R}^4$, $m = 1, 2, 3, 4$ are columns of $\mathbf{A}$. It is known that $\text{rank}(\mathbf{A}) = 2$ and $\mathbf{C}_2 = 3\mathbf{C}_1$ and $\mathbf{C}_4 = 2\mathbf{C}_1 + 3\mathbf{C}_3$. If a particular solution of $\mathbf{Ax} = \mathbf{b}$ is $[1, 0, 1, 0]^T$, then

(a) Find the general solution of $\mathbf{Ax} = \mathbf{b}$.

(b) Find $\mathbf{b}$ if $\text{RREF}(\mathbf{A}) = \mathbf{A}$.

SOLUTION



(a) Now $\mathbf{A} = [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4]$ where $\mathbf{C}_m \in \mathbb{R}^4$, $m = 1, 2, 3, 4$ are columns of $\mathbf{A}$. It is known that $\text{rank}(\mathbf{A}) = 2$

$$\mathbf{C}_2 = 3\mathbf{C}_1 \Rightarrow 3\mathbf{C}_1 - \mathbf{C}_2 + 0\mathbf{C}_3 + 0\mathbf{C}_4 = \mathbf{0}$$

$$\Rightarrow [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4] \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}$$

$$\mathbf{C}_4 = 2\mathbf{C}_1 + 3\mathbf{C}_3 \Rightarrow 2\mathbf{C}_1 + 0\mathbf{C}_2 + 3\mathbf{C}_3 - 1\mathbf{C}_4 = \mathbf{0}$$

$$\Rightarrow [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4] \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} = \mathbf{0}$$

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Thus, the general solution of $\mathbf{Ax} = \mathbf{b}$ is given by

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} \mid \forall \lambda_1, \lambda_2 \in \mathbb{R} \right\}$$

(b) Now $\mathbf{A} = [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4] = [\mathbf{C}_1, 2\mathbf{C}_1, \mathbf{C}_3, 2\mathbf{C}_1 + 3\mathbf{C}_3]$

and $\text{rank}(\mathbf{A}) = 2$.

Therefore $\mathbf{C}_1, \mathbf{C}_3$ are pivot columns and $\text{RREF}(\mathbf{A}) = \mathbf{A}$

$$\Rightarrow \mathbf{C}_1 = [1, 0, 0, 0]^T \text{ and } \mathbf{C}_3 = [0, 1, 0, 0]^T$$

$$\Rightarrow \mathbf{C}_2 = [3, 0, 0, 0]^T \text{ and } \mathbf{C}_4 = [2, 3, 0, 0]^T$$

$$\Rightarrow \mathbf{A} = \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \text{ Therefore } \mathbf{b} = \mathbf{A}[1, 0, 1, 0]^T = [1, 1, 0, 0]^T$$

PROBLEM 4

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Consider $M = \{\mathbf{A} \in \mathbb{R}^{2 \times 2} \mid \mathbf{A} = -\mathbf{A}^T\}$.

- (a) Prove that M is a subspace of vector space V , where V is set of all 2×2 real matrices.
- (b) Prove or disprove $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ is a basis for M defined in above.
- (c) Prove or disprove that $\left\{ \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 5 & -5 \\ 5 & 5 \end{bmatrix}, \begin{bmatrix} 0 & -3 \\ 3 & 1 \end{bmatrix} \right\}$ is a linearly independent set.

SOLUTION



(a) Clearly the zero matrix of order 2, $\mathbf{0} \in M \Rightarrow M \neq \phi$.

Let $\mathbf{A}, \mathbf{B} \in M, k \in \mathbb{R}$

$$\Rightarrow \mathbf{A} = -\mathbf{A}^T, \mathbf{B} = -\mathbf{B}^T$$

$$\Rightarrow \mathbf{A} + \mathbf{B} = -\mathbf{A}^T - \mathbf{B}^T = -(\mathbf{A} + \mathbf{B})^T$$

$$\Rightarrow \mathbf{A} + \mathbf{B} \in M$$

$$k\mathbf{A} = k(-\mathbf{A}^T) = (-k)\mathbf{A}^T = -(k\mathbf{A})^T$$

So, M is a subspace.

(b) Clearly $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ does not belong to M and hence cannot be a basis of M .

SOLUTION



(c) Consider $a \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} + b \begin{bmatrix} 5 & -5 \\ 5 & 5 \end{bmatrix} + c \begin{bmatrix} 0 & -3 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$$\Rightarrow a + 5b = 0$$

$$-a - 5b - 3c = 0$$

$$a + 5b + 3c = 0$$

$$5b + c = 0$$

$\Rightarrow a = b = c = 0$. Hence it is a linearly independent set.

PROBLEM 5



Given two linearly independent $D \times 1$ vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, where $D > 3$, we intend to create a $D \times 3$ matrix where each column of the matrix is a linear combination of the given vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. If possible, find vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ and values for the coefficients in the linear combination such that the resulting matrix has full column rank. Otherwise, explain with a suitable mathematical argument why this is not possible.

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Let us take the first column of the matrix to be $\mathbf{p} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2$, and the second column to be $\mathbf{q} = d_1 \mathbf{v}_1 + d_2 \mathbf{v}_2$.

Since our goal is to ensure that the resulting matrix have rank 3, we must ensure that the first two columns are linearly independent.

This is achieved when $c_1 d_2 \neq c_2 d_1$.

In order to achieve full column rank (i.e. rank = number of columns), we need to ensure that the third column of the matrix cannot be written in terms of the first two columns, i.e. $\mathbf{p}$ and $\mathbf{q}$.

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However the third column of the matrix is given to be some linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$. If we are able to write $\mathbf{v}_1$ and $\mathbf{v}_2$ in terms of $\mathbf{p}$ and $\mathbf{q}$, then the third column can be written in terms of $\mathbf{p}$ and $\mathbf{q}$ making the third column a linear combination of the first two, which means less than full column rank. To see that the third column can be written in terms of the first two, consider the following equations:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 = \mathbf{p}$$

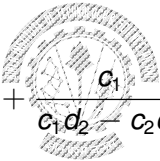
$$d_1 \mathbf{v}_1 + d_2 \mathbf{v}_2 = \mathbf{q}$$

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$$\mathbf{v}_2 = -\frac{d_1}{c_1 d_2 - c_2 d_1} \mathbf{p} + \frac{c_1}{c_1 d_2 - c_2 d_1} \mathbf{q}, (\because c_1 d_2 \neq c_2 d_1)$$

$$\mathbf{v}_1 = -\frac{d_2}{c_2 d_1 - c_1 d_2} \mathbf{p} + \frac{c_2}{c_2 d_1 - c_1 d_2} \mathbf{q}, (\because c_1 d_2 \neq c_2 d_1)$$

INNER PRODUCT



- For a real-valued, finite-dimensional vector space V and an ordered basis B of V , it holds that $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is an inner product if and only if there exists a symmetric, positive definite matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ with

$$\langle \mathbf{x}, \mathbf{y} \rangle = \hat{\mathbf{x}}^T \mathbf{A} \hat{\mathbf{y}}$$

- Inner products and norms are closely related in the sense that any inner product induces a norm:

$$\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}.$$

- Inner product induced norm satisfies Cauchy Schwartz inequality: $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$ and angle between $\mathbf{x}, \mathbf{y}$ is given by $\cos^{-1} \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}$

PROBLEM 6

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$$\text{Let } \mathbf{A} = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

Then define $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = \mathbf{x}^T \mathbf{A} \mathbf{y}$ where $\mathbf{x}, \mathbf{y} \in \mathbb{R}^4$.

(a) Prove that $\langle \cdot, \cdot \rangle_{\mathbf{A}}$ is an inner product on $\mathbb{R}^4$.

(b) Find all possible $\mathbf{y} = [y_1, y_2, y_3, y_4]^T$ such that $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = 0$ where $\mathbf{x} = [1, 0, 0, 0]^T$

SOLUTION



(a) Clearly $\mathbf{A} = \mathbf{A}^T$ and hence $\mathbf{A}$ is symmetric. Now

$$\begin{aligned}\mathbf{x}^T \mathbf{A} \mathbf{x} &= 2x_1^2 - 2x_1x_2 + 2x_2^2 - 2x_2x_3 + 2x_3^2 - 2x_3x_4 + 2x_4^2 \\ &= x_1^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + (x_3 - x_4)^2 + x_4^2 > 0\end{aligned}$$

if $\mathbf{x} \neq \mathbf{0}$.

Thus $\mathbf{A}$ is symmetric positive definite. Hence by theorem $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = \mathbf{x}^T \mathbf{A} \mathbf{y}$ is an inner product.

(b) If $\mathbf{y} = [y_1, y_2, y_3, y_4]^T$ and $\mathbf{x} = [1, 0, 0, 0]^T$

we want $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = 0$

$$\Rightarrow 2y_1 - y_2 = 0 \Rightarrow 2y_1 = y_2$$

Thus $\mathbf{y} = [y_1, y_2, y_3, y_4]^T$ with $2y_1 = y_2$, $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = 0$.

PROBLEM 7

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Using the Cauchy–Schwarz inequality, show that

$$(a_1 + a_2 + \cdots + a_n)^2 \leq n(a_1^2 + a_2^2 + \cdots + a_n^2)$$

for arbitrary real numbers $a_1, a_2, \dots, a_n$. Do there exist values of $a_1, a_2, \dots, a_n$ for which equality holds?

SOLUTION



Consider the application of the Cauchy-Schwartz inequality on two vectors:

$\mathbf{a} = (a_1, a_2, \dots, a_n)^T$ and $\mathbf{b} = (1, 1, \dots, 1)^T$.

Then the Cauchy-Schwartz inequality gives $|\mathbf{a}^T \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$.

We get,

$$|(a_1 + a_2 + \dots + a_n)| \leq \sqrt{(a_1^2 + a_2^2 + \dots + a_n^2)} \sqrt{n}$$

Squaring both sides, we get the required result.

If we substitute $a_i = 1$, $i = 1, \dots, n$, the inequality turns into an equality.

PROBLEM 8

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Consider the positive definite matrix $A \in \mathbb{R}^{3 \times 3}$ defined as:

$$A = \begin{bmatrix} 9 & 6 & 3 \\ 6 & 8 & 12 \\ 3 & 12 & 75 \end{bmatrix}$$

- (i) Derive a lower triangular matrix L so that A can be factorized as $A = LL^T$. Derive all the three eigenvalues of L .
- (ii) Let $x \in \mathbb{R}^3$ be a nonzero vector. Define $z = L^T x$ where $L \in \mathbb{R}^{3 \times 3}$ satisfies the equality $A = LL^T$. Prove that z can never be the zero vector.

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Given

$$A = \begin{bmatrix} 9 & 6 & 3 \\ 6 & 8 & 12 \\ 3 & 12 & 75 \end{bmatrix}$$

Since A is positive definite, it admits a Cholesky factorization $A = LL^T$ where L is lower triangular.

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

Then

$$LL^T = \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{31}l_{11} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

SOLUTION



Equating with A:

$$l_{11}^2 = 9 \Rightarrow l_{11} = 3$$

$$3l_{21} = 6 \Rightarrow l_{21} = 2$$

$$3l_{31} = 3 \Rightarrow l_{31} = 1$$

$$l_{21}^2 + l_{22}^2 = 8 \Rightarrow 4 + l_{22}^2 = 8 \Rightarrow l_{22} = 2$$

$$l_{21}l_{31} + l_{22}l_{32} = 12 \Rightarrow 2 + 2l_{32} = 12 \Rightarrow l_{32} = 5$$

$$l_{31}^2 + l_{32}^2 + l_{33}^2 = 75 \Rightarrow 1 + 25 + l_{33}^2 = 75 \Rightarrow l_{33} = 7$$

$$L = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 2 & 0 \\ 1 & 5 & 7 \end{bmatrix}$$

SOLUTION



Since L is triangular, its eigenvalues are its diagonal entries:

$$\lambda_1 = 3, \quad \lambda_2 = 2, \quad \lambda_3 = 7$$

(ii) Let $x \in \mathbb{R}^3$ be nonzero and define $z = L^T x$. Then

$$x^T A x = x^T L L^T x = (L^T x)^T (L^T x) = z^T z = \|z\|^2$$

Since A is positive definite,

$$x^T A x > 0 \quad \forall x \neq 0$$

Hence

$$\|z\|^2 > 0 \Rightarrow z \neq 0$$

$z = L^T x$ can never be the zero vector if $x \neq 0$.

PROBLEM 9



- (a) Assume that you are given a matrix $A \in \mathbb{R}^{n \times n}$ where $A = [a_{ij}]$ ($1 \leq i, j \leq n$). The matrix A has the following interesting property:

$$a_{i1} + a_{i2} + \cdots + a_{in} = 0 \quad \text{for each of } i = 1, 2, \dots, n.$$

- (i) Can zero be one of the eigenvalues of this matrix? Give reasoning for your answer.
- (ii) Derive the determinant of this matrix.
- (iii) Derive at least one eigenvector–eigenvalue pair of this matrix.

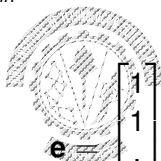
SOLUTION



(a) Given $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ such that

$$a_{i1} + a_{i2} + \cdots + a_{in} = 0 \quad \text{for each } i = 1, 2, \dots, n.$$

(i) Let


$$\mathbf{e} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

Then

$$A\mathbf{e} = \begin{bmatrix} a_{11} + a_{12} + \cdots + a_{1n} \\ a_{21} + a_{22} + \cdots + a_{2n} \\ \vdots \\ a_{n1} + a_{n2} + \cdots + a_{nn} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

SOLUTION



Hence

$$A\mathbf{e} = 0\mathbf{e}.$$

0 is an eigenvalue of A .

(ii) Since 0 is an eigenvalue, A is singular. Therefore,

$$\det(A) = 0.$$

(iii) An eigenvalue–eigenvector pair is

$$\lambda = 0, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

PROBLEM 9



- (b) Consider a vector $r \in \mathbb{R}^k$. Here $k > 132$. Consider a matrix M defined as follows: $M = rr^T$. Based on this definition answer the following:
- (i) How many non-zero eigenvalues does the matrix M have? Derive all the nonzero eigenvalues.
 - (ii) Find the sum of absolute value of eigenvalues of this matrix i.e.

$$\sum_{i=1}^k |\lambda_i|$$

where λ_i is the i^{th} eigenvalue of M .

SOLUTION



(b) Let $r \in \mathbb{R}^k$ with $k > 132$ and define

$$M = rr^T$$

(i) The matrix $M = rr^T$ is of order $k \times k$, and

$$\text{rank}(M) = 1 \quad (\text{if } r \neq 0).$$

Hence, M has exactly one nonzero eigenvalue.

Now,

$$Mr = rr^T r = r(r^T r) = (r^T r)r.$$

SOLUTION



Thus r is an eigenvector with eigenvalue

$$\lambda_1 = r^T r = \|r\|^2.$$

All other eigenvalues are zero.

(ii) The eigenvalues of M are

$$\|r\|^2, 0, 0, \dots, 0.$$

Therefore,

$$\sum_{i=1}^k |\lambda_i| = \|r\|^2.$$



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Thank you